

DISASTER MANAGEMENT

IMPORTANT CONCEPTS AND TERMINOLOGIES

MODULE - 1



LOOK AT THE FOLLOWING PICTURES!



CYCLONE KYARR



EARTHQUAKE - PHILIPPINES



KAVALAPPARA LANDSLIDE



AMAZON RAINFOREST FIRE



TORNADO



VOLCANO ERUPTION

HOW CAN WE ADDRESS THEM?



HOW TO DEFINE A DISASTER?



It is a serious disruption of the functioning of a society, causing widespread human, material or environmental losses which **exceed the ability of the affected society to cope using its own resources.**

COPING CAPACITY

It is the ability of people, organizations and systems to use available skills and resources to manage adverse conditions, risk or disasters.



HAZARD

A natural /man made threat that is present in a certain region

RISK

Possibility and extent of a hazard, usually calculated and can be minimised

EXPOSURE

How much the human life and related objects situated near the hazard



DISASTER

The event that happens, extraordinary emergency interventions are required



VULNERABILITY

Quantitatively correlated with risk, identifies the damageability

CAPACITY

Reciprocal to vulnerability, tries to minimise the risk

MITIGATION

Some actions taken to help reduce or eliminate long-term risks



13 OCTOBER

INTERNATIONAL DAY
FOR DISASTER RISK REDUCTION

DISASTER RISK REDUCTION

Systematic development and application of policies, strategies and practices to minimize vulnerabilities



WHAT ARE HAZARDS?

- 1 A hazard is a **situation where there is a threat to life**, health, environment or property. For eg: If a city is built over a active seismic zone or a coastal city that is highly prone to tsunamis



WHAT ARE DISASTERS?

- 1 A disaster is **an event that completely disrupts the normal ways of a community**. It brings on human, economical, and environmental losses to the community which the community cannot bear on its own.



WHAT ARE HAZARDS?

- 2** Hazards occur at the place which has less population. A massive volcano in an uninhabited island won't be called a disaster



WHAT ARE DISASTERS?

- 2** Disasters mainly occur at over populated areas. A massive volcano in a highly populated city or near it will be called a catastrophic disaster



WHAT ARE HAZARDS?

- 3** Hazard is caused by negligence. Eg: a chemical factory can cause occupational and environmental hazards without proper safety measures.



WHAT ARE DISASTERS?

- 3** Disasters are results of differential behaviour of nature due to many conditions. Eg: Due to global warming and stalling of winds, India will be prone to many catastrophic cyclones



WHAT ARE HAZARDS?

- 4 Hazards are **natural or manmade phenomenon** that are a feature of our planet and cannot be prevented. In their dormant state, they just pose a threat to life and property.



WHAT ARE DISASTERS?

- 4 The **hazards turn into disasters when they cause widespread destruction** of property and human lives. It is the phase when the event has actually occurred and recorded.



DISASTER MANAGEMENT

TYPES OF HAZARDS

NATURAL



MAN-MADE



SOCIO-NATURAL



DISASTER MANAGEMENT

TYPES OF HAZARDS

NATURAL



- Caused by the forces of nature and man has no role to play in such hazards.
- Following are the main examples of hazards
 - Earthquakes
 - Volcanic eruptions
 - Cyclonic storms
 - Tsunamis
 - Floods
 - Droughts
 - Landslides



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TYPES OF HAZARDS

MAN-MADE



Caused by undesirable activities of man

- Explosions
- Leakage of toxic waste
- Pollution of air, water and land
- Dam failures
- War and Civil strife
- Terrorism



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TYPES OF HAZARDS

SOCIO-NATURAL

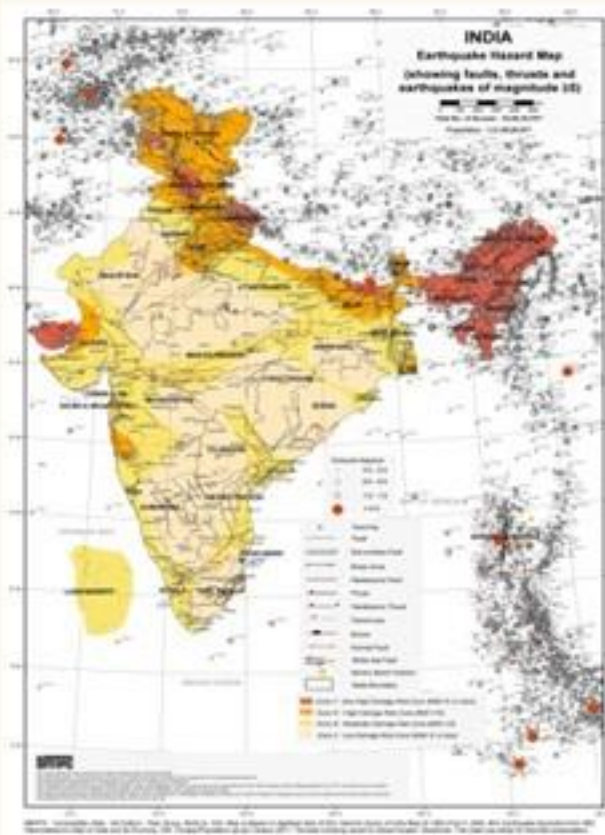


Caused by the combined effect of natural forces and misdeeds of man.

- Eg 1; the frequency and intensity of floods and droughts may increase due to indiscriminate felling of trees, particularly in the catchment areas of the rivers.
- Eg 2: Storm surge hazards may be worsened by the destruction of mangroves.
- Eg 3: Although landslides are normally caused by natural forces, yet they may be caused and their frequency and impact may get aggravated as a result of building roads in mountainous areas, excavating tunnels and by mining and quarrying.



HAZARD ZONATION MAPPING



WHAT ARE THEY?

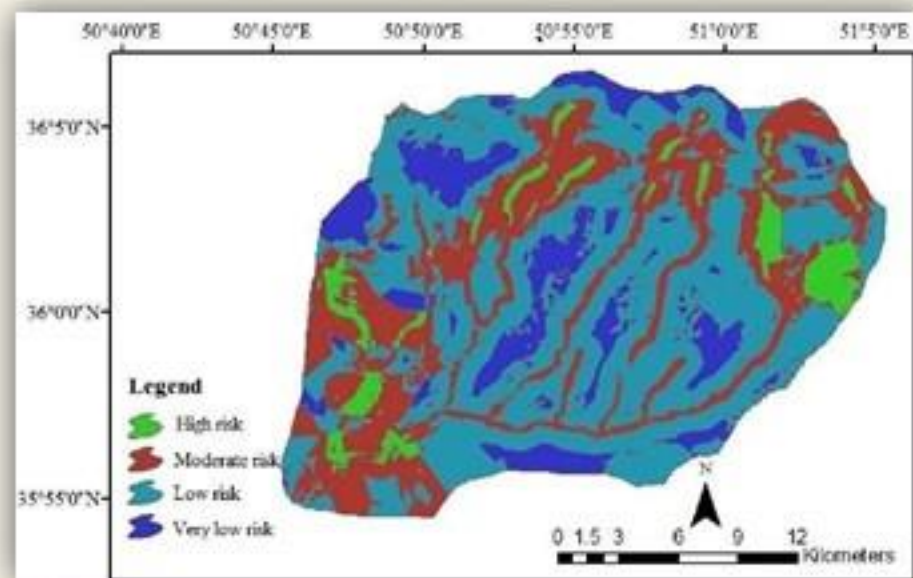
Hazard maps provide

- Details of extent of risk involved due to hazards along with details of extent of geographical impact;
- Level of risk and geographical details like extent of human settlement;
- Provides data about resources and infrastructure that will be affected.

*The Indian subcontinent is among the world's most disaster prone areas.
Almost 85% of India's area is vulnerable to one or multiple hazard*



HAZARD ZONATION MAPPING

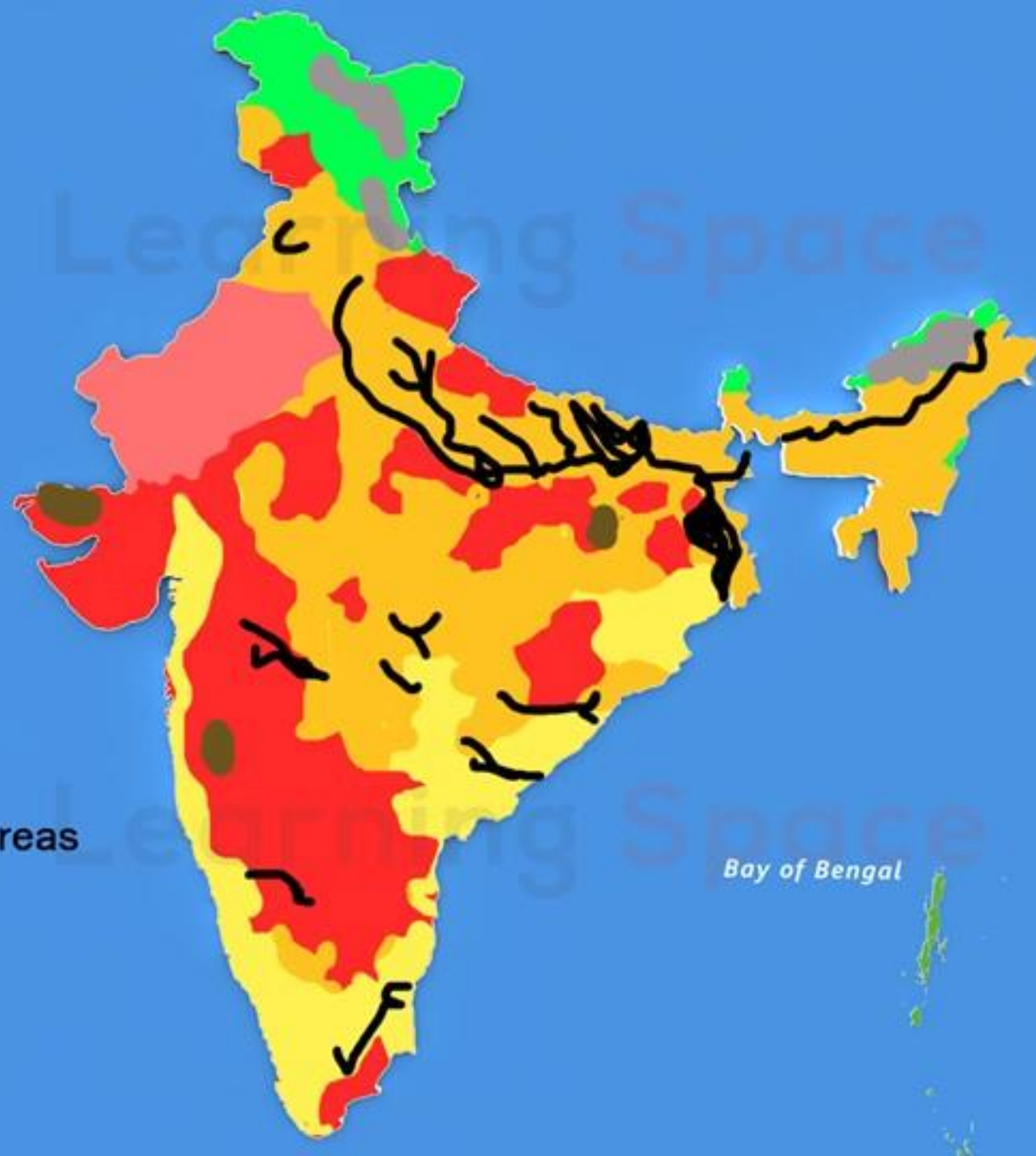


WHY ARE THEY USEFUL?

- 1 A study area or a sub-area can be properly assessed. Study areas can be divided into sub-areas requiring more information, additional assessments, or specific reduction techniques.
- 2 A more realistic evaluation of risks to new development is possible. Appropriate hazard reduction techniques can be more easily developed to deal with an emergency situation.
- 3 Rehabilitation can be initiated in advance with proper coordination among different central and state agencies.

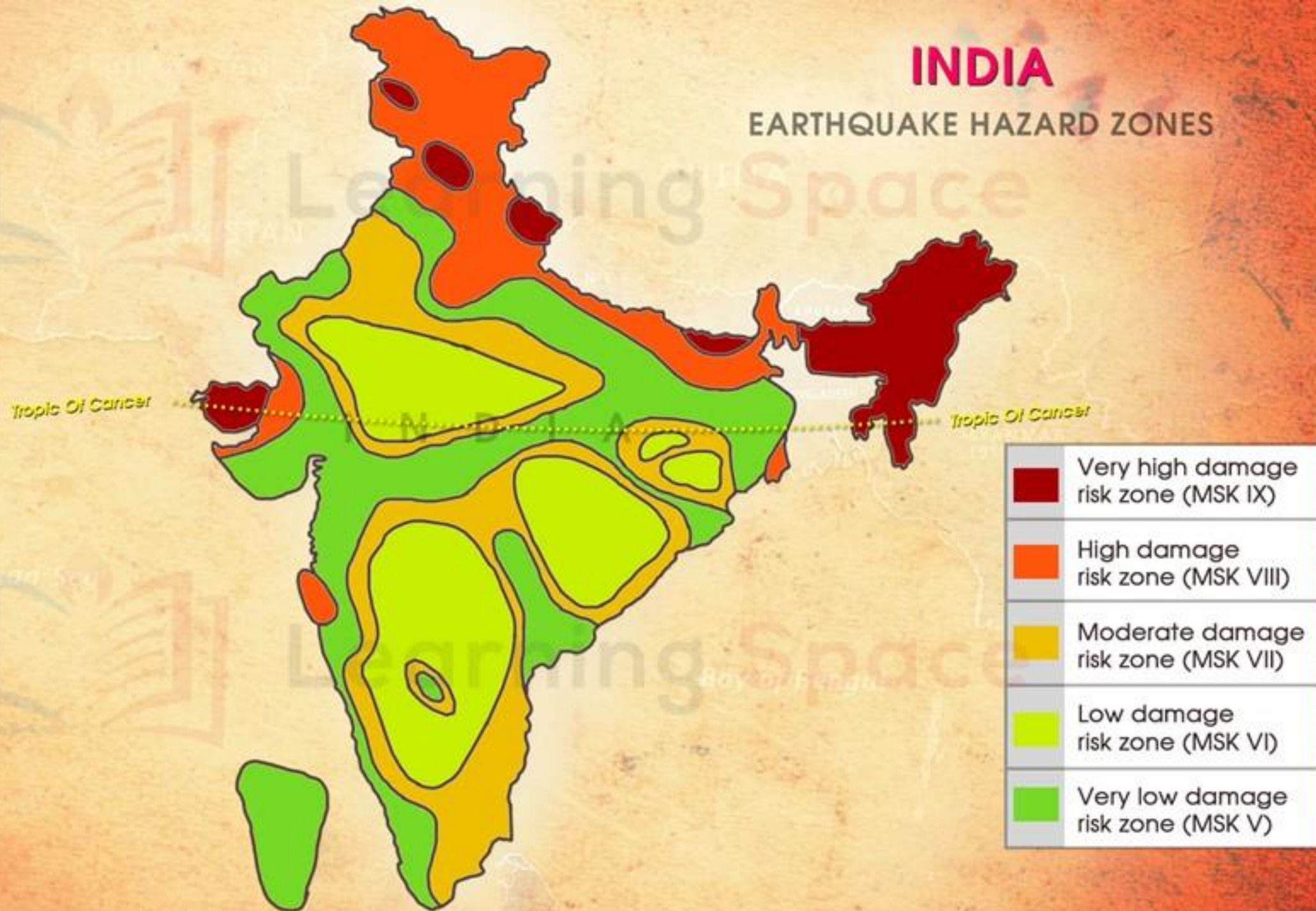


-  Drought prone
-  Cyclone prone areas
-  Hot Desert
-  Cold Desert
-  Floods
-  Earthquakes

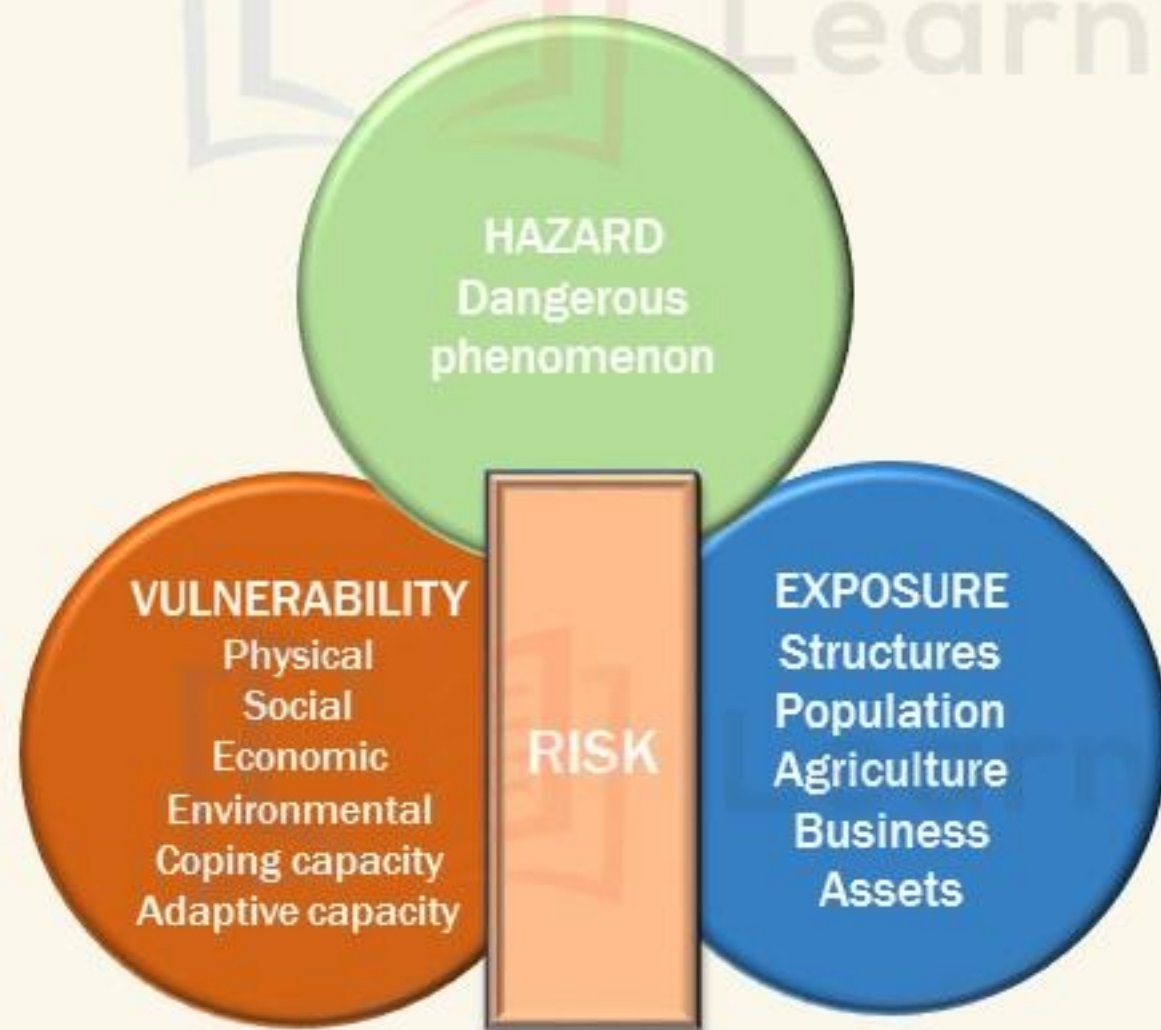


INDIA

EARTHQUAKE HAZARD ZONES



EVALUATING RISK OF A DISASTER



- Disaster Risk assessment is a important process to determine the nature and extent of any impending disaster.
- Risk assessments can quantitatively analyze hazards and evaluate existing conditions of vulnerability
- The assessments are helpful in predicting what and how much could potentially harm exposed people, property, services, livelihoods and the environment.

Also check: Global Earthquake Model
Global Risk Data Platform



HOW TO MEASURE RISK FOR BETTER MANAGEMENT?

LOSS ACCOUNTING

HOW MUCH WAS LOST?

- Recording of impact
- Measuring trends



- Local Policy
- National Policy
- International Policy

DISASTER FORENSIC

HOW MUCH WAS LOST IN ONE EVENT?

- Identifying the cause
- Learning from past events



- Local expert teams
- National expert teams
- International Expert teams

RISK MODELLING

HOW MUCH WE LIKELY TO LOSE IN FUTURE?

- Modelling future losses
- DRR and Mitigation



- Local research/ policy
- National research/policy
- International initiatives

As an officer, you can take any event like a heatwave or an earthquake and calculate the following to identify the loss and potential threats in your area.



LOOK OUT FOR THE FOLLOWING WHEN YOU ARE INVOLVED IN RISK MANAGEMENT





RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE



- 1** Disaster management is more than just reaction to crises, it is a holistic process that needs a fresh look in light of frequent natural calamities. Examine
- 2** How important are vulnerability and risk assessment for pre-disaster management. As an administrator, what are key areas that you would focus in a disaster management. (UPSC 2013)
- 3** What is hazard mapping and why is it necessary for framing policies suitable to cope with disasters? How are they related to reducing disaster risk?
- 4** Disasters are no longer seen only as unfortunate one-off events to be responded to, but also as deep-rooted and longer-term problems that must be planned for. Justify the statement with suitable examples.



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